

NUMERICAL INTEGRATION

Trapezoidal rule

Let the interval $[a, b]$ be divided into n equal sub intervals such that $a = x_0 < x_1 < x_2 < \dots < x_n = b$. Clearly, $x_n = x_0 + nh$. Hence the integral becomes

$$I = \int_{x_0}^{x_n} y dx$$

Approximating y by Newton's forward difference formula, we obtain

$$I = \int_{x_0}^{x_n} \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{6}\Delta^3 y_0 + \dots \right] dx$$

Since $x = x_0 + ph$, $dx = hdp$ and hence the above integral becomes

$$I = h \int_0^n \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{6}\Delta^3 y_0 + \dots \right] dp$$

which gives on simplification

$$\int_{x_0}^{x_n} y dx = nh \left[y_0 + \frac{n}{2}\Delta y_0 + \frac{n(2n-3)}{12}\Delta^2 y_0 + \frac{n(n-2)^2}{24}\Delta^3 y_0 + \dots \right] \dots\dots\dots(1)$$

From this general formula, we can obtain different integration formulae by putting $n = 1, 2, 3, \dots$, etc.

Setting $n = 1$ in the above general formula, all differences higher than the first will become zero and we obtain

$$\int_{x_0}^{x_1} y dx = h \left(y_0 + \frac{1}{2}\Delta y_0 \right) = h \left[y_0 + \frac{1}{2}(y_1 - y_0) \right] = \frac{h}{2}(y_0 + y_1)$$

For the next interval $[x_1, x_2]$, we deduce similarly

$$\int_{x_1}^{x_2} y dx = \frac{h}{2}(y_1 + y_2)$$

and so on. For the last interval $[x_{n-1}, x_n]$, we have

$$\int_{x_{n-1}}^{x_n} y dx = \frac{h}{2}(y_{n-1} + y_n)$$

Combining all these expressions, we obtain the rule

$$\int_{x_0}^{x_n} y dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n]$$

which is known as the trapezoidal rule.

The error of the trapezoidal formula can be obtained in the following way. Let $y = f(x)$ be continuous, well-behaved, and possess continuous derivatives in $[x_0, x_n]$. Expanding y in a Taylor's series around $x = x_0$, we obtain

$$\int_{x_0}^{x_1} y dx = \int_{x_0}^{x_1} \left[y_0 + (x - x_0)y'_0 + \frac{(x - x_0)^2}{2}y''_0 + \cdots \right] dx = hy_0 + \frac{h^2}{2}y'_0 + \frac{h^3}{6}y''_0 + \cdots$$

Similarly,

$$\begin{aligned} \frac{h}{2}(y_0 + y_1) &= \frac{h}{2} \left(y_0 + y_0 + hy'_0 + \frac{h^2}{2}y''_0 + \frac{h^3}{6}y'''_0 + \cdots \right) \\ &= hy_0 + \frac{h^2}{2}y'_0 + \frac{h^3}{4}y''_0 + \cdots \end{aligned}$$

So we obtain

$$\int_{x_0}^{x_1} y dx - \frac{h}{2}(y_0 + y_1) = -\frac{1}{12}h^3y''_0 + \cdots,$$

which is the error in the interval $[x_0, x_1]$. Proceeding in a similar manner we obtain the errors in the remaining sub intervals, viz., $[x_1, x_2]$, $[x_2, x_3]$, $\dots$ and $[x_{n-1}, x_n]$. We thus have

$$E = -\frac{1}{12}h^3 (y''_0 + y''_1 + \cdots + y''_{n-1}),$$

where E is the total error.

since $nh = b - a$